



Arsh Dhawan

Indian Institute of Science Education and Research Bhopal

Phone No.: +91 9466472600

E-mail: arsh23@iiserb.ac.in

BS-MS

in LinkedIn

Website

EDUCATION

- **BS-MS**
Physical Sciences
IISER Bhopal, India
2023 - Present
- **Higher Secondary / +2**
Adarsh Jain, Delhi
2023
- **Secondary**
Scholars Rosary, Rohtak
2021

About

I am a fourth-year Physics Student interested in pursuing Theoretical Physics, especially interested in pursuing QFT and Mathematical Physics in future. I find both physics and mathematics fascinating especially topics like GR, QFT and Group Theory, and try to approach them by their most basic postulates. My experience under Prof. Ganguli has made me more inclined to understand the subject by focusing on essence of the subject, which translates better in Physics and Mathematics than in any other field. Moreover the opportunity under Dr. Naga Deepthi made me familiar with current advancements in Neutrino Physics, like 'Supernova Neutrinos', 'SNSI' and 'Decoherence', while Dr. Rahul introduced me to Dark Matter Particle and its Phenomenology via projects on 'Dark Matter Particle' and 'Scotogenic Model',

I am also familiar with a few specialized coding packages like GLoBES, SNoWGLoBES, S.A.R.A.H., SPheno, Madgraph and MicroOmegas; and proficient in languages like Mathematica and MATLAB. Though, I am familiar I wish not to work in coding heavy projects and am more inclined to projects which include mathematics/theory and phenomenology or/and model building. I am willing to do some coding for realization of models as I did in Scotogenic pipeline but am not a fan of all-code projects.

Experiences

- **Research Project: Scotogenic model for Dark Matter particle and its computational implementation** Dr. Rahul Srivastava
In this project, we aim to understand Scotogenic model for Dark Matter, which beautifully connects with Neutrino Physics specially the mass mechanism for Neutrinos using See-Saw mechanism. For this model we shall be considering neutrinos as majorana fermions having minuscule masses. This model produces 1 fermionic and 1 bosonic particles which can be considered as contenders for Dark Matter.
In addition to this, we shall be working upon implementing this model, first in S.A.R.A.H., then making our way to SPheno, finally implementing and deriving results from MicroOmegas and Madgraph.
- **Summer Project: Using Higgs-like induced gravity model for Gravitational Baryogenesis** Dr. David Silva Pereira
In this project, I explored the origin of cosmic matter through the lens of modified-gravity thermodynamics by taking GR as an equilibrium limit of Scalar-tensor theories. In scalar-tensor baryogenesis, evolving scalar can generate a chemical potential that favors matter over antimatter. We connected these two pictures by expressing the baryogenesis source in terms of the effective thermodynamic variables of scalar-tensor gravity. I specifically used Non-minimally Coupled Higgs-like Model / Induced-Gravity Limit and tried to assess whether this offers a clear and consistent interpretation of matter generation in modified gravity.
- **Summer Internship 2026; Decoherence in Neutrino Scalar Non-standard Interactions in oscillation dynamics** Dr. Deepthi K. N.
Under Dr. Deepthi, I am continuing my work on Scalar Non Standard Interactions and Neutrino Oscillations under Scalar NSI. We are hopeful for a publishable result and are almost near the end of this work
I was working on a mixing matrix approach to Scalar non-standard interactions(Snsi) for which I found effective mixing matrices and tried finding analytical expression which currently seem unfeasible considering exponentially increasing number of terms with increasing order as we attempted using all Snsi parameters at a time which hasnt been done as of now. This approach provided an easy way into OQS and decoherence study in Snsi and especially focused on DUNE experiment. Our formulation can also be extended to include Vector NSI elements too. You can request relevant pdf if necessary. We are hoping to upload an arXiv preprint soon.

- Non-Hermitian Dynamics in 2-flavour Neutrino Oscillations(Summer 2026)** Dr. Poonam Mehta
 Under Dr. Poonam Mehta I am working on a project which as planned would attempt in studying Non Hermitian Physics starting from OQS, and better understanding Entanglement and Bell Theorem.
 Following this we shall attempt to use G-metric and Density Matrix approach in Non-Hermitian Physics for 2-flavour neutrino-oscillations. This project is heavily influenced by Dr. Poonam's recent paper on 'Two flavor neutrino oscillations in presence of non-Hermitian Dynamics'.
- Project: Dark Matter and Its Contenders(2025-26)** Dr. Rahul Srivastava
 During winter(Dec-Mar'26), I took up a project under Dr. Rahul Srivastava, Associate Professor, IISER Bhopal where I approached Dark Matter Phenomenology using Particle Physics(BSM). For this I referred to Dr. Gianfranco Bertone white paper and book on the same alongside other resources. Other than this I also worked on limiting cases of neutrino physics under constraints imposed by solar, atmospheric and Supernova neutrinos.
- ICTP PWF Riemannian and Complex Geometry (2025-26)** ICTP
 ICTP offers a select few students, courses in a few topics every year. During Sep'25-Feb'26, I was part of one such course focused upon "Riemannian and Complex Geometry" which introduced me to topology, differential, riemannian and complex geometry along with some concepts/practical applications in General Relativity.
- Teaching Assistant, Quantum Physics (2025)** Dr. Dr.Nirmal Ganguli
 Under Prof. Ganguli, I acted as TA for Quantum Physics course, helping other students understand the topics better by solving their doubts and sometimes, teaching a few during the Tutorials.
- Summer Internship 2025; Supernovae Neutrinos and Neutrino Oscillations** Dr. N. K. Deepthi
 Under Dr. Naga Deepthi, I worked in the field of Neutrino Oscillations covering the MSW effect in matter. Later, I focused on the study of supernova and the neutrino emission during such astrophysical events. I was also introduced to the fundamental concepts of Open Quantum Systems as well as basic QFT and Particle Physics, required for the project. I also delved into Computational aspect, making use of GLOBES to understand neutrino phenomenology better. Later following the collaboration, I studied Amol Dighe's paper on 'Identifying FLux from Supernova Explosions' as well as Kuo and Mikheyev's paper on Resonance in Neutrino Oscillations.
- Reading Project; Quantum Mechanics** Dr. Arnab Rudra
 Under Dr. Rudra, in an extended program, he encouraged learning quantum mechanics by a problem solving approach, making us familiar with basics of QM as well as Dirac notation.
- Summer School; General Relativity and Cosmology** Dr. Shibesh Kumar Jas, Director, PICS
 Under Dr. Shibesh, I got familiar with General Relativity and Black Hole Thermodynamics, using computational Tools like Mathematica throughout this project.

Working Paper

- Probing the combined effects of scalar non-standard interactions and environmental decoherence at DUNE**

Abstract:

The Deep Underground Neutrino Experiment (DUNE) is an upcoming long-baseline neutrino oscillation experiment with the potential to probe subleading effects including scalar non-standard interactions (SNSI), and environmental decoherence. The study of these new physics scenarios provides an opportunity to explore physics beyond the Standard Model (BSM).

In this work, we investigate the combined effects of SNSI and environmental decoherence on neutrino oscillation probabilities. We study how these two effects modify the sensitivity of DUNE to neutrino mass hierarchy (MH) determination and CP violation. Furthermore, we explore the sensitivity of DUNE to distinguish between the signatures of vector NSI and scalar NSI in a dissipative environment.

We are thinking of presenting it at DAE-BRNS High Energy Physics (DAE-BRNS-HEP) Symposium-2026 as a poster.

TECHNICAL SKILLS

- **Kernels-** Linux
- **Programming Languages-** C
- **Basic/Simple Programming Languages-** MATLAB, Mathematica
- **Technical Writing-** LaTeX
- **Neutrino Specific packages-** GLoBES, SNowGLoBES
- **HEP packages:** S.A.R.A.H., SPheno, MicroOmegas, Madgraph

Relevant Courses

- **Mechanics**
B+
- **Quantum Physics**
A
- **Calculus**
B
- **Groups and Symmetries**
B
- **Quantum Mechanics-I**
B
- **Standard Model of Particle Physics**
A [Audited]

Knowledge from other Sources

- **Quantum Information Theory**
- **Open Quantum Systems**

Extra-Curricular Activities

- **Playwright** 2023-24
AARAMBH [IISER-B Drama Club]
- **Head Writer** 2024-2025
AARAMBH[IISER-B Drama Club]
- **Content writer** 2023-24
Singularity Tech Fest [IISER-B]
- **Content writer** 2023-24
Mahuaa Fest [IISER-B]

Volunteer

- **Literary Fest, Arts and Letters** 2024
- **Integration Bee** 2024
Acted as event organizer of "Integration Bee" for IISER-B Math Fest, Continuum.